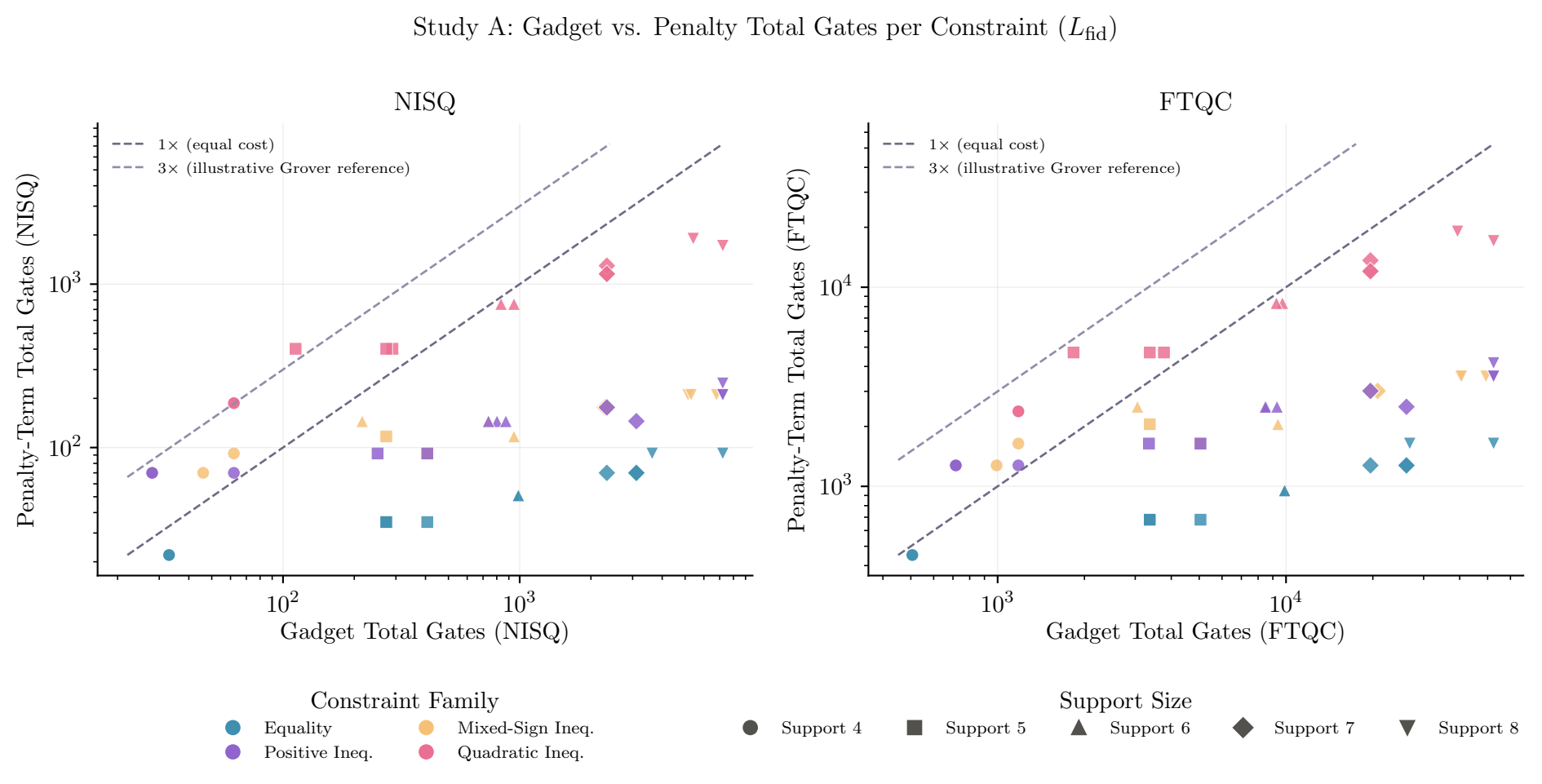

Study A: Gadget vs. Penalty Total Gates per Constraint (L_{fid})
NISQ
FTQC
1× (equal cost)
3× (illustrative Grover reference)
Penalty-Term Total Gates (NISQ)
Gadget Total Gates (NISQ)
Penalty-Term Total Gates (FTQC)
Gadget Total Gates (FTQC)
Constraint Family
Equality
Positive Ineq.
Mixed-Sign Ineq.
Quadratic Ineq.
Support Size
Support 4
Support 5
Support 6
Support 7
Support 8